



DRAFT

SOUTHERN AFRICAN DEVELOPMENT COMMUNITY (SADC)

CYBER-INFRASTRUCTURE FRAMEWORK

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Executive Summary

The purpose of this SADC Cyber-Infrastructure Framework is to form a basis for a planned SADC Cyber-Infrastructure Strategy and Action Plan. It also proposes an implementation plan and governance structure for such a Cyber-Infrastructure strategy.

A SADC Cyber-Infrastructure Strategy and Action Plan will support the implementation of the approved SADC Strategic Plan on Science, Technology and Innovation (STI) and also contribute concretely to the Research and Development, Innovation and Industrialisation pillar of the Digital SADC 2027. The SADC Strategic Plan on STI is a five year plan (2015-2020) to guide the region to create an enabling environment to harness science, technology and innovation as a tool to address socio-economic challenges for sustainable development in the region. This Framework has been developed to also support the implementation of the Protocol on Science, Technology and Innovation; Revised Regional Indicative Strategic Development Plan (RISDP), Industrialization Strategy and Roadmap; and Regional Infrastructure Development Master Plan (RIDMP).

Prior to this the SADC Cyber-Infrastructure Framework proposal, a draft SADC HPC (High Performance Computing) Framework had been proposed. That framework considered regional Infrastructure development, Research and Development; Human Capital Development; Governance and strategic partnerships around High Performance Computing. However, with the contemporary rapid escalation in the importance of Data, collaborative research, data sharing; the growing dependence on broadband connectivity and developments in regional national integrated Cyberinfrastructure Systems, the focus shifted towards the development of this holistic, integrated SADC Cyber-Infrastructure Framework.

1. Introduction

Investment in the areas of science, technology and engineering including the establishment and support of research institutions and provision of technological support creates a country-level climate of innovation. These types of investment foster knowledge creation and dissemination and have a strong positive influence on the long-term competitiveness of a country. Research and innovation in all domains is provably becoming increasingly cyber-infrastructure dependent.

The Southern African Development Community (SADC) Declaration on Information and Communications Technology (ICT) indicates that the region will endeavour to develop a national information and communication infrastructure. The proposed Cyber-Infrastructure Framework would provide a coherent roadmap to enhance and develop a holistic Cyber-Infrastructure System in the SADC region to:

- impact socio-economic development and industrialisation;
- enhance education, not just educate the people in developing technologies but to use them as well; and
- support collaborative research development and innovation.

This Framework utilises the following working definition of Cyber-Infrastructure (CI):

Cyber-Infrastructure (CI) is a key driver for a knowledge based economy- it comprises technologies, skills, people and policies which support the generation, analysis, transport and sharing as well as stewardship of information.

This Framework, in itself, does not offer a Cyber-Infrastructure Strategy, but rather motivates for and provides a Roadmap towards developing such a Strategy.

Figure 1 illuminates the breadth and depth of a well-developed Cyber-Infrastructure ecosystem.

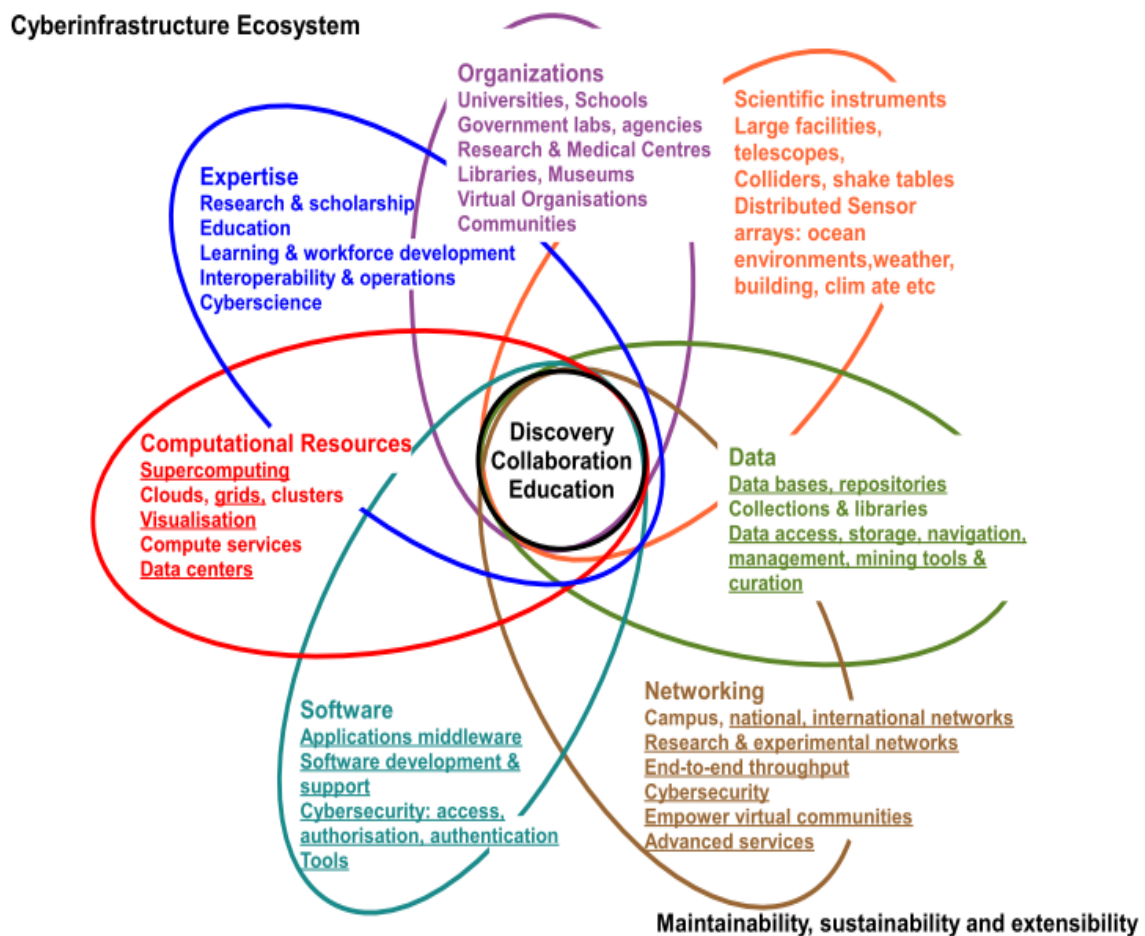


Figure 1: View of a well-developed Cyber-Infrastructure Ecosystem

There are many examples of individual countries and regions adopting economic and developmental strategies to enhance their socio-economic stature. The European Union (EU) has adopted a "Digital Single Market" strategy empowered by cyber-infrastructures, which the EU estimates will contribute 145 billion euros per annum to the EU economy and create "hundreds of thousands on new jobs". The UK has created a UK Data Forum to "provide strategic direction to the ongoing development and maintenance of a coherent and

robust national data structure for UK social and economic research and research in related fields of interest". An academic study has concluded that the contribution of "research data curation and sharing might be worth at least AUS\$1.8 billion and possibly up to AUS\$5.5 billion a year". These are but three examples.

2. Background: Impact of Cyber-Infrastructures

Modern Cyber-Infrastructures (termed e-Infrastructures in some parts of the world) are the bedrock of the digital age. A coherent systemic world-class Cyber-Infrastructure provides the platform for a nation or region (e.g. EU) to advance its digital economy. In the SADC region it would enable moving from resource based to knowledge based economies by virtue of its impact on:

- E-education;
- E-health;
- E-government; and
- E-agriculture.

A further benefit would be its Impact on Regional Integration as a consequence of:

- Collaboration using cyber-infrastructure;
- Utilising CI to encourage collaborations in other sectors (e.g. energy, education, health, agriculture);
- Poverty alleviation;
- Impact on Industrialisation, namely so-called Industry 4.0; and
- Technology transfer (spin-outs/start-ups, incubators), commercialisation and industrialisation as a consequence of research and education advances.

An effective Cyber-Infrastructure system could have a spectrum of potential further consequences, for example:

- Citizen science, i.e. contributions to research by citizens, has become feasible because of powerful data (which includes video and audio) data gathering devices and networks; citizen science has become an important tool in encouraging research amongst scholars and in classrooms and even gathering data about endangered ecosystems;
- Digital and nationally shared information repositories are replacing traditional libraries;
- Disciplines previously untouched by the digital age are impacted by eScience / eResearch;
- Social media, e.g., texting, tweating, and pinpointing the location (of individuals or groups, vehicles, etc.) have relevance to the social sciences, disaster response systems, security systems, geophysical monitoring etc.;
- Sharing data, advancing to open data and open research better enables multi-disciplinary and multi-national type research than the previous closed ("it is my data!") paradigm; and

- The rising tide of non-reproducible research can be largely countered by enforcing open data and open research principles.

All of these developments require a comprehensive cyberinfrastructure strategy.

3. Cyber-Infrastructures

Components of a Cyber-Infrastructure include:

- **Research and Education Networks (RENs)** - specialised broadband infrastructure networks and service providers that exclusively support research, education and innovation communities.
- **Computation resources**- ranging from HPC to other computing capabilities and services.
- **Data**- Implementation of tools and facilities to enable efficient data driven discoveries, technologies and innovations.
- **HR capacity development** to enable:
 - CI specialists to roll-out services and infrastructure;
 - Beneficiaries to fully benefit from the services provided by CI's.
- **Policies** –to enable optimum establishment and utilisation of cyber-infrastructure.

4. Situational Analyses of existing Cyber-Infrastructures within SADC Member States

Cyber-Infrastructure activity has been evident within member countries over the past few years. These activities will be clustered by major Cyber-Infrastructure type.

4.1. Situational Analysis for High Performance Computing (HPC)

Prior to the technical meeting in (2011), a survey was developed and circulated to all Member States in order to enquire about the HPC situation in each state and to determine which industries would be receptive to HPC utilisation. The survey was also meant to ascertain whether HPC facilities were available to Member State's research communities. The survey highlighted numerous industries within the SADC region that require HPC. Pertinent sectors include: engineering, education, energy; socio-economic issues pertaining to indigenous medicine, climate change, health care, water provision, weather forecasting, mining and knowledge sharing.

The survey at that stage was focused only on HPC and did not cover other cyber-infrastructure components (i.e. RENs and research data infrastructure).

The region has already forged strategic partnerships with international HPC institutions such as Texas Advanced Computing Center (TACC) University of Texas, International Center for Theoretical Physics, STEM TREK and others to facilitate acquisition of HPC equipment's and its deployment to SADC Member States for initial HPC systems

There are currently existing HPC systems in the SADC region which are at various operational levels, for example in states such as Botswana, South Africa, Lesotho, Mozambique, Tanzania, Zambia and Zimbabwe which other Member States can access.

A summary of responses received from Member States appears below:

- **Botswana:** TACC HPC Ranger racks received by University of Botswana, Computer Science Department. The system is currently used to support HPC and Data science capacity building and will support existing projects in Faculty of Science and Engineering. There are possible further HPC projects in the pipeline. Challenges and needs include; funding, joint research projects and further Human Capital Development programmes to evolve and grow this service to support more stakeholders
- **Lesotho:** HPC user platform and HPC system from the Japanese government for Lesotho Meteorological Services. Main users: Meteorological services, climate analysis and Mets services. There is a need for regional centres to facilitate access sharing, collaboration and capacity building programmes.
- **Mozambique:** Government Data Centre. Planned projects include; e-Government research, SADC SKA collaboration. Challenges cited; low legal validity of electronic transactions
- **South Africa:** Centre for High Performance Computing (CHPC) operational since 2007, supports different areas of research applications. A new HPC Petaflop machine was installed in 2016 and ranks at #121 among the Top500 supercomputers, CHPC has more than 300 active users.
- **Tanzania:** PARAM Serengeti operational since 2010 and two Ranger Racks deployed. Main users; weather information management, usage model-system by Masters and PhDs students. Challenges with power and cooling, lack of awareness and understanding of HPC, weak linkages with industry.
- **Zambia:** Received 2 Ranger Racks in early 2015 (hosted by University of Zambia) which were deployed in August 2015. Focus areas include; environmental sciences, education, health and mining. Designated host for Astronomy Office for Southern Region. Needs cited; awareness creation, funding and sustainability and regional strategy.
- **Zimbabwe:** Zimbabwe Centre for High Performance Computing (ZCHP) operational since 2015. Has users from life sciences; weather and climate forecasting and academic research. Challenges encountered are financial and to deal with different users. There is also a need for supporting infrastructure and technical capacity.

4.2. Situational Analysis for National Research and Education Networks (NRENs)

The Ubuntunet Alliance is the internationally recognised Regional Research and Education Network for Southern and Eastern Africa. Regional RENS (RRENs) provide support to NRENs (National Research and Education Networks) in establishing their in-country infrastructure and services capabilities, acting as a bulk-buying agent to acquire services on

behalf of member NRENs, as well as assisting with the provisioning of infrastructure to allow member NRENs to inter-connect to one another, as well as to NRENs globally.

The SADC region falls within the region supported and covered by the UbuntuNet Alliance. SADC Member States that are currently members of the UbuntuNet Alliance include DRC, Madagascar, Malawi, Mozambique, Namibia, South Africa, Tanzania and Zambia. UbuntuNet also has member NRENs in countries outside SADC, such as the mature NREN of Kenya, viz KENET.

With respect to SADC area NRENs: there are currently two mature NRENs, namely the South African NREN (combined capabilities of SANReN and TENET), and the Zambian NREN, ZAMREN.

Several fledgling NRENs are currently evolving within the SADC region, e.g. MoREnet (Mozambique) and TERNET in Tanzania. UbuntuNet, as well as mature NRENs, for example the SA NREN, ZAMREN and KENET, are playing an important support role in assisting these nascent NRENs.

4.3. Situational Analysis for Data

There are a number of developments in SADC Member States with regard to data and information dissemination, for example e-government projects, data centers, open data and open government initiatives:

- President Kikwete of Tanzania (September 2015): *“For us, Open Data is the way to go if we are to transform our country from Least Developed Country to a Middle Income Country, and ultimately to become an advanced economy. We are looking at the future of Open Data in Tanzania with much optimism, there is no reason or intentions of back tracking. You can count on us.”*
- As a component of the South Africa’s National Integrated Cyber-infrastructure System (NICIS) ; Data Intensive Research Initiative for South Africa (DIRISA) is being developed.
- The Open Data for Africa platform is a response from the African Development Bank Group (AfDB) aimed at boosting access to quality data necessary for managing and monitoring development results in African countries, including the African Action Plan 2063 and (SDGs) Sustainable Development Goals.
- The Africa Data Consensus is a strategy for implementing the data revolution in Africa that aims to create a new statistical landscape, opening up the field of data production and dissemination to state and non-state actors.

Adopted in March 2015 at the High Level Conference on the Data Revolution, in response to calls for a framework on the data revolution in Africa and its implications for the African Union’s (AU) Agenda 2063 and the Sustainable Development Goals (SDGs). The plan of action will be guided by the United Nations Economic Commission for Africa (UNECA); the African Union Commission (AUC) and the African Development Bank (AfDB); with support from the United Nations Development Programme (UNDP); and the United Nations Population Fund

(UNFPA), and implemented in collaboration with partner institutions from the public and private sectors as well as civil society organisations.

4.4. Alignment of Cyber-Infrastructure Framework with SADC

This proposed SADC Cyber-Infrastructure Framework supports the implementation of the approved SADC Strategic Plan on Science, Technology and Innovation (STI). The SADC Strategic Plan on STI is a five year plan (2015-2020) to guide the region to create an enabling environment to harness science, technology and innovation as a tool to address socio-economic challenges for sustainable development in the region. The Framework has been developed to support the implementation of the Protocol on Science, Technology and Innovation; Revised Regional Indicative Strategic Development Plan (RISDP), Industrialization Strategy; and Regional Infrastructure Development Master Plan (RIDMP). It should be noted that Cyber-infrastructure falls within Research, Innovation and Industry Development Pillar of the Digital SADC 2027 which is the ICT Chapter of RIDMP.

5. Background to Framework

In August 2009, the Blue Gene for Africa initiative—presented to the SADC senior officials for ScienceTechnology and Innovation meeting by the South African Centre for High Performance Computing (CHPC)—initiated the development of the SADC HPC Platform which led to a process of developing a HPC Strategy / Framework.

Follow up HPC technical meetings were held in 2011, 2012, and 2013, when the focus was put on the three areas:

- Pilot projects on areas of economic activity that can be enhanced by cyber-infrastructure (HPC);
- Strategy of developing human capital resources for the efficient utilisation of HPC resources; and
- Regional R&D strategy to address regional priority challenges.

The SADC Joint Ministerial meeting on Science, Technology and Innovation, Education and Training held in Maputo, Mozambique in June 2014, noted that a draft HPC Framework has been developed to guide the implementation and roll-out of HPC initiatives in the SADC region and also guide Member States in crafting their own national policies and strategies. The impact of data and broadband developments on education, research and innovation has resulted in the need to evolve the earlier SADC HPC Framework into a broader SADC Cyber-infrastructure Framework.

5.1. RATIONALE AND POLICY FRAMEWORKS

This proposed SADC Cyber-Infrastructure Framework is consistent with existing SADC Policy Frameworks, for example:

- AU Agenda 2063
- STISA 2024
- SADC Treaty
- SADC Protocol in Science, Technology and Innovation
- RISDP Revised 2015
- SADC Strategic Plan on STI (2015-2020)
- RIDMP-Digital SADC 2027
- SADC Industrialization Strategy (2015-2063)

5.2. Rationale for this Framework

The SADC mission statement is "to promote sustainable and equitable economic growth and socio-economic development through efficient productive systems, deeper co-operation and integration, good governance, and durable peace and security, so that the region emerges as a competitive and effective player in international relations and the world economy". In relation to this mission, the region recognises the importance of ICT as a cross-cutting enabler in addressing the challenges posed by globalisation, facilitating the regional integration agenda, and enhancing the socio-economic and political development prospects of the region. Cyber-infrastructure enables cutting-edge research and development within universities, research institutions as well as industry.

The purpose of this Framework is to create awareness and to guide the development of a SADC Cyber-Infrastructure Strategy and an Action Plan. Further, it is also to promote the development of National Cyber-Infrastructure policies and strategies by each of the Member States. The Framework provides for:

- Cyber-Infrastructure Development;
- Governance and Institutional Arrangements;
- Education, Research & Development and Innovation; and
- Strategic Partnerships.

The Framework calls for a subsequent implementation plan that will identify projects and activities and also address resource mobilisation.

5.3. Strategic Goals and Objectives of Framework

The goal is that this Framework will seed and form a basis for a CI Strategy Plan to:

- Promote high-level quality education, research and innovation;
- Accelerate Industrialization and commercialisation in SADC; and
- Promote shared Cyber-Infrastructure commons (infrastructure and capabilities).

The ultimate Goals are that it will:

- Add value to scientific programmes;
- Foster partnerships and collaborations;
- Develop regional Cyber-Infrastructure networks through interconnect, HPC centres and Research & Scientists; and
- Enhance the region's human capacities.

6. Cyber-Infrastructure: Strategic Goals and Objectives

The vision is an education, research, and innovation environment that provides for human capital development and shared access to unique or distributed facilities regardless of location to impact socio-economic development and industrialisation in the SADC region and promote knowledge based economy.

Strategic Goals are to:

- Promote high quality education, research and innovation;
- Build cyber-infrastructure capacity;
- Promote cyber-infrastructure commons (an environment to share education, research and innovation resources); and
- Accelerate industrialisation and commercialisation in SADC.

7. FOCUS AREAS

These Focus Areas need to be unpacked in order to develop a comprehensive SADC Cyber-Infrastructure Strategy:

1. Policy/Strategy development, institutionalisation, implementation and support;
2. Education, Research & Development and Innovation;
3. Human Capital Development;
4. Infrastructure development;
5. Resource mobilisation;
6. Communication, Awareness & Advocacy; and
7. Strategic Partnerships.

7.1. Policy/Strategy development, institutionalisation, implementation and support

This Framework provides guidelines which will assist Member States in crafting their own national cyber-infrastructure policy frameworks. It is imperative for Member States to note that the Framework is to be utilised as template to encourage policy formulation and implementation.

- SADC to develop harmonised SADC Model Policy to be institutionalised by Member States, to create a conducive environment for the promotion of cyber-infrastructure;
- Work towards regulations that enable open access to commercial infrastructure, such as Preferential pricing from commercial infrastructure providers;
- Strategy and Action Plan for SADC Cyber-Infrastructure, encompassing regional projects; and
- Regular monitoring and evaluation reporting.

7.2. Education, Research & Development and Innovation

- Cyber-Infrastructure support for education;
- Support existing and new Centres of Excellence (CoE) and provide them with tools;
- Cyber-Infrastructure support for research by promoting research collaborations and supporting regional flagship projects; and
- Cyber-Infrastructure support for innovation, for example via technology transfer (spin-outs/start-ups, incubators), commercialisation and industrialisation.

7.3. Human Capital Development

Human Capital Development is an important factor for the CI initiative within the region. The introduction of CI into the mainstream academic system should be prioritized; and facilitated at a national level through the establishment of an academic curriculum. According to the SADC Declaration on ICT the region undertakes to formulate a comprehensive human resource development plan in the area of ICT to enable the Region to meet its needs and participate as an equal partner in the new technologically informed global economy. The effectiveness use of ICT presumes a literate population which is able to use the tools provided by new technologies.

- Create a pool of CI experts.
- Train beneficiaries to fully exploit the services provided by a) Incorporating mainstream elements of CI in higher education curriculum development, and b) promoting e-Readiness to beneficiaries.
- Collaborations with other regional and international education and research institutions and partners.

- Collaboration with industry with respect to a) Vendor industries with regard to co-operation on human capital development and b) Consumer industries to promote utilisation of CI

7.4. Infrastructure development

- Overarching issues: Infrastructure sustainability through lifecycle management.
- Creation of a cyber-infrastructure commons; adhering to international standards, policies, etc.
- HPC: Development of adequate regional computational capacity; Awareness programs on HPC and other computing modalities.
- Networking: Establishment of NRENs and regional interconnectivity; Access and participate in AfricaConnect Projects and UbuntuNet Alliance.
- Data: Establish trusted data repositories that are regionally inter-federated; internationally benchmarked policies and standards for data stewardship- open access, sharing and interoperability.

7.5. Strategic Partnerships

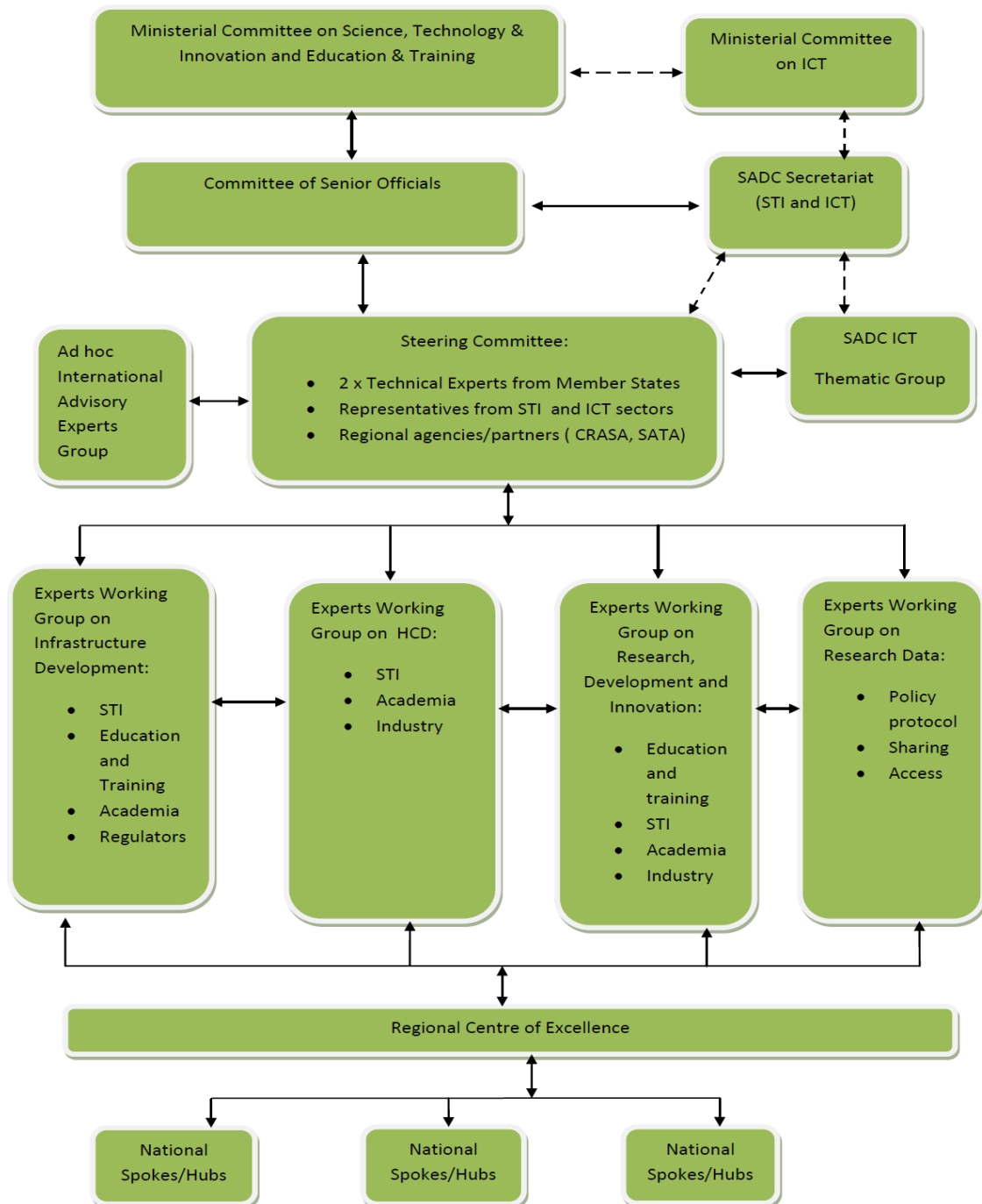
- The region has forged strategic partnerships with international Cyber-Infrastructure institutions such as Texas Advanced Computing Centre (TACC), International Centre for Theoretical Physics (ICTP), STEM-Trek to facilitate acquisition of Cyber-Infrastructure equipment such as the “Ranger Racks” which have been deployed to institutions in Botswana, South Africa; Tanzania and Zambia.
- Strategic partnerships will be forged with multilateral; private; and public institutions to support implementation of the Regional Cyber-Infrastructure Framework. Partners will be engaged to invest in implementation of Cyber-Infrastructure capacity building; and research, development and innovation programmes. The Secretariat will facilitate the establishment of Public Private Partnerships to invest in the roll out of Cyber-Infrastructure equipment acquisition in the region.

8. Implementation Mechanism

The implementation mechanism of the Cyber-Infrastructure Framework includes governance and institutional arrangements. The implementation of the Framework will be driven by the SADC Secretariat in partnership with SADC Member States; regional and international partners. The implementation of the Framework will be aligned to the general principles of the revised RISDP (2015-2020), SADC Industrialization Strategy and Roadmap (2015-2063), Regional Infrastructure Development Master Plan (RIDMP), Protocol on Science, Technology and Innovation and other relevant protocols and strategies. The key stakeholders in the implementation of the Cyber-Infrastructure Framework would include the following:

- (i) SADC Ministerial Committee on Science, Technology and Innovation and Education and Training;
- (ii) SADC Ministerial Committee on ICT;
- (iii) Committee of Senior Officials;
- (iv) SADC Secretariat through the STI Unit and ICT Unit;
- (v) Regional Cyber-Infrastructure Steering Committee;
- (vi) SADC ICT Thematic Group;
- (vii) Ad hoc International Advisory Experts Group;
- (viii) Experts Working Group on Infrastructure Development;
- (ix) Experts Working Group on Human Capital Development (HCD);
- (x) Experts Working Group on Research, Development and Innovation;
- (xi) Experts Working Group on Research Data;
- (xii) Regional Centres of Excellence; and
- (xiii) National Spokes/Hubs

Figure 2: Institutional framework for implementation of the SADC Cyber-Infrastructure Framework



9. Governance Structure

The governance structure will serve as an implementing mechanism to guide the implementation of the Framework. The governance structure will provide policy guidance; set standards and priorities to facilitate implementation of the Framework.

9.1. The SADC Ministerial Committee on Science, Technology and Innovation and Education and Training

The Ministerial Committee will include Ministers responsible for Science, Technology and Innovation and Education and Training. The role of the Ministerial Committee will provide policy guidance and approve policy and programmes of Senior Officials in relation to the implementation of the Cyber-Infrastructure Framework. The Ministerial Committee will consider and approve programmes and projects in relation to the following strategic focus areas:

- Policy development, implementation and support;
- Research, development and innovation;
- Research Data;
- Human Capital Development;
- Infrastructure Development and Investment; and
- Communications, awareness and advocacy.

9.2. Ministries of Information and Communication Technology

The Ministries of Information and Communication Technology will be responsible for providing ICT policy guidance in terms of regulations, broadband infrastructure and connectivity.

9.3. Committee of Senior Officials

The Committee of Senior Officials will be responsible to coordinate and direct the implementation of the Framework. The Committee will include representatives from the following sectors; STI, Education & Training and ICT. The Committee will consider policy, programmes and projects recommendations from the Cyber-Infrastructure Steering Committee on the implementation of the Framework.

9.4. SADC Secretariat (STI and ICT Unit)

The SADC Secretariat through the STI and ICT Units will facilitate the execution and monitor and evaluate the implementation of the Cyber-Infrastructure Framework. The SADC Secretariat will perform a secretarial and coordination role in the governance structure. The

Secretariat will also be responsible for tabling progress reports on the implementation of the Framework to the Ministerial Committees through the Senior Officials Committee.

9.5. Member States

The obligation of the Member States will be to provide leadership, establish supportive administrative mechanisms, and formulate, monitor and evaluate the Cyber-Infrastructure Framework. Member States will also be responsible for developing national strategies, strengthening institutions and providing funding for implementation of the Framework.

9.6. Regional Cyber-Infrastructure Steering Committee

A Regional Cyber-Infrastructure Steering Committee will be established. The Steering Committee will consist of technical experts from SADC Member States representing the STI, Education and Training and ICT sectors. The Steering Committee will also consist of regional agencies and partners from STI and ICT sectors. This Steering Committee will be responsible for the coordination, facilitation, monitoring and evaluation of the implementation of the SADC Cyber-Infrastructure Framework. The Steering Committee will consider and approve programmes and projects from Technical Working Groups. The Steering Committee will also be responsible for resource mobilisation and leveraging of strategic partnerships to support the implementation of the Framework.

9.7. SADC ICT Thematic Group

The SADC ICT Thematic Group will comprise of International Cooperating Partners (ICPs), regional stakeholders and SADC STI and ICT implementing agencies. The role of the SADC ICT Thematic Group is to facilitate resource mobilisation and consider programmes and projects for implementation of the Cyber-Infrastructure Framework. The SADC ICT Thematic Group will also act as a platform for dialogue as well as the interface between SADC and ICPs to improve the coordination of all stakeholders.

9.8. Ad hoc International Advisory Experts Group

An Ad hoc International Advisory Experts Group on Cyber-Infrastructure will be established. The Ad hoc International Advisory Experts Group will consist of Experts drawn from Cyber-Infrastructure Institutions and Organisations of international stature, covering strategy and policy as well as a practitioner focus across all cyber-infrastructure elements. This Ad hoc International Advisory Experts Group will provide technical and policy advice to the Steering Committee.

9.9. Experts Working Groups

The following Experts Working Groups will be established:

9.9.1. Experts Working Group on Infrastructure Development and Investment

The Experts Working Group on Infrastructure Development and Investment will consist of representatives from sectors such as; STI, Education and Training, Academia, ICT (ministries, regulators and operators). The role of the Experts Working Group on Infrastructure Development and investment will among others include the:

- Identification of infrastructure needs at national and regional level;
- Development of proposals to fund and maintain infrastructure; and
- Establishment and deployment of cyber infrastructure at national and regional level.

9.9.2. Experts Working Group on Human Capital Development (HCD)

The Experts Working Group on Human Capital Development will consist of representatives from sectors such as STI, Education, Academia, ICT and Industry. The role of the Experts Working Group on HCD amongst others will include the:

- Identification of human capital development needs and priorities;
- Conceptualisation of capacity building initiatives and programmes; and
- Leveraging of external partners and funding for human capital development programmes in Cyber-Infrastructure.

9.9.3. Experts Working Group on Research, Development and Innovation

The Experts Working Group on Research, Development and Innovation will consist of representatives from sectors such as Education and Training, STI, Academia and Industry. The role of the Experts Working Group on Research, Development and Innovation will include amongst others the:

- Identification of research needs and priorities;
- Development of regional research agenda on Cyber-Infrastructure;
- Establishment of regional research consortia;
- Development of research proposals; and
- Leveraging of partnerships and funding for conducting research.

9.9.4. Experts Working Group on Research Data

The Experts Working Group on Research Data will consist of technical experts on research data. The role of the Experts Working Group on Research Data will be to develop framework and guidelines to facilitate data stewardship, access and storage and other data challenges across Member States. The Experts Working Group will also be responsible for issues related to Intellectual Property Rights (IPR).

9.10. Regional Centres of Excellence

The Regional Centres of Excellence will be identified and strengthened to serve as the platform for the implementation of the Cyber-Infrastructure Framework. The Centres of Excellence will; conduct research, innovation, capacity building & training; and facilitate technology transfer.

9.11. National Spokes/Hubs

The National Spokes/Hubs will be identified by Member States to coordinate and implement the regional Cyber-Infrastructure Framework. These will include tertiary institutions, science councils, research councils, public (including parastatals) and private institutions.

10. Communication, Awareness and Advocacy

A stakeholder engagement plan will be formulated to clearly stipulate various methods that can be used to engage with different stakeholders. These will include a number of instruments such as STI weeks, outreach programmes that can be used for advocacy along with the use of interactive and informative road shows. In addition to these, a Cyber-infrastructure website or portal will be established for information dissemination and knowledge sharing. The outreach activities will assist to showcase cyber-infrastructure capabilities and benefits.

A Regional Cyber-Infrastructure awareness, communications and advocacy strategy will be developed. The communication, awareness and advocacy activities will include engagements with sectors such as; ICT, health, education both at public and private sector level.

11. Funding and Resources

Cyber-infrastructure as a science and research enterprise requires substantive investment, but the social and economic returns are high in the long run. Operational, maintenance and sustainability of these infrastructures require dedicated funding and commitment from relevant stakeholders in both the public and private sectors. The SADC Secretariat will mobilise resources from international cooperating partners, Member States, and private sector to provide both financial and technical resources for the implementation of the Framework. Other funding instruments such as the Regional Development Fund and European Development Fund will also be leveraged to support the implementation of the Framework.

The following should be considered:

- Formulating development of a resource and mobilisation plan;
- Leverage funds to support activities for cyber-infrastructure development and promotion;
- Approach vendors and industry for funding; and

- Capitalise on grants and calls, which constitute part of the Bilateral Funding.

12.Strategic Partnerships with Vendors and Industry

The region has forged strategic partnerships with international Cyber-Infrastructure institutions such as Texas Advanced Computing Centre (TACC), International Centre for Theoretical Physics (ICTP), STEM-Trek to facilitate acquisition of Cyber-Infrastructure equipment such as the “Ranger Racks” which have been deployed to institutions in Botswana, South Africa; Tanzania and Zambia.

Strategic partnerships will be forged with multilateral; private; and public institutions to support implementation of the Regional Cyber-Infrastructure Framework. Partners will be engaged to invest in implementation of Cyber-Infrastructure capacity building; and research, development and innovation programmes. The Secretariat will facilitate the establishment of Public Private Partnerships to invest in the roll out of Cyber-Infrastructure equipment acquisition in the region.

13.Monitoring and Evaluation, Reporting and Learning

The monitoring and evaluation of the Framework and the strategic plan will be guided by the SADC Policy on Strategy Development, Planning, Monitoring and Evaluation (SPME) which was approved by SADC Council of Ministers in 2012. The Framework will be subjected to a Mid-Term Review to ensure the strategic focus areas and deliverables are effectively and efficiently implemented. The Framework will be monitored and evaluated by the SADC Secretariat and Cyber-Infrastructure Steering Committee.

14. Conclusion

This document has provided a SADC Cyber-Infrastructure Framework as a precursor to the development of a Strategy and Action plan. The framework highlights aspects of the proposed cyberinfrastructure; provides alignment with existing SADC policy frameworks and initiatives, and it proposes implementation priorities and a governance structure. Areas of importance to developing a Cyber-Infrastructure Strategy are highlighted. A regional cyberinfrastructure will contribute concretely to Education, Research & Development and Innovation and advance the SADC regional objectives around STI and accelerate regional industrialisation.

Cyber-infrastructure is favourably positioned to make a positive contribution to the growth of most knowledge intensification strategies in existing and new industries. It forms one of the pillars and facilitators of national and global research and innovation. As such it is an important contributor underpinning strategies aimed at growing the quality and quantity of scientific and engineering endeavours in the region. It plays a key role in the

internationalisation of a country or region's scientific and engineering efforts as well as the social, economic, and health sciences.